

Sugarcane — Crop Production Notes

Introduction and Local Names

The scientific name of sugarcane is *Saccharum officinarum* L., and it belongs to the family Poaceae. In Urdu, sugarcane is called Naishkar, and in Punjabi it is known as Ganna or Kamad.

Species of Sugarcane

Saccharum officinarum is known as American Cane, *Saccharum sinense* as Chinese Cane, *Saccharum barberi* as Indian Cane, *Saccharum spontaneum* as Wild Asian Cane, and *Saccharum robustum* as the Wild Cane of New Guinea.

Crop Characteristics

Sugarcane is a perennial, C4 crop with a determinate growth habit. It is cross-pollinated, and it is propagated using fuzz (true seed) or setts (stem cuttings).

Composition

Sugarcane is composed of 11–16% fibre, 12–16% soluble sugars, 2–3% non-sugars, and 63–73% water.

Top Sugarcane Producing Countries

According to FAO (2019) data, Brazil led sugarcane production with 37,300,000 tonnes, followed by India with 26,605,000 tonnes, China with 11,474,000 tonnes, Thailand with 10,024,000 tonnes, and the USA with 7,666,000 tonnes.

Sugarcane Production in Pakistan (2019)

In 2019, sugarcane was cultivated on an area of 1,270,000 hectares in Pakistan, with a total production of 83,000,000 tonnes and a yield of 65.3 tonnes per hectare. Pakistan ranked 6th in world sugarcane production that year.

Importance

Sugarcane is primarily grown for sugar extraction. Its bagasse is used to make chip board, paper, and hard board. Its molasses are used for alcohol production, and its tops and trash serve as fodder for animals. Sugarcane is also processed locally into gur, shakar, and desi cheeni.

Climate Requirements

Temperatures above 20°C are suitable for sugarcane growth, while temperatures below 20°C stunt its growth. The crop thrives in hot, humid conditions typical of tropical and subtropical areas, requires ample sunlight, and goes through a dormant period during winter.

Soil Requirements

Sugarcane grows best in well-drained soils ranging from heavy to loam, with high organic matter content. Sandy soils can be made suitable by adding organic matter. The crop is sensitive to saline soils.

Root Development

Sugarcane has three types of roots. Primary or sett roots are shorter in length and shorter-lived, and are responsible for initial growth; they originate from the sett bands and are also called first roots. Secondary, tiller, or stem roots are longer in length and longer-lived, providing original growth; these develop after tillers form and provide nutrition and anchorage to the plant. Adventitious or fibrous roots can extend up to 2 metres.

Shoot and Cane Development

The sugarcane stem is jointed, with pithy internodes that may be thick, medium, or thin. Internodes are smaller at the base and top of the cane, while the internodes in the mid part are larger, though this can be affected by water stress or high temperature.

Cane color can be white, pinkish, or greenish. Internode shapes vary and include cylindrical, tumescent, bobbin, conoidal or abconoidal, and curved forms.

Leaf

Sugarcane leaf blades vary in size and shape, and may be parallel, semi-erect, or drooping. Thick leaves tend to be erect or semi-erect, while thin leaves tend to be drooping or horizontal. Soft, green leaves are favored by animals as fodder. The top leaves remain active, while dry leaves are known as trash or khori.

Inflorescence

The sugarcane inflorescence is arrow-shaped and takes the form of a spike, resembling Sarkanda. It develops under conditions of high temperature and humidity, and is notably found in areas such as Murree and Thattah (Sajawal).

Primary Tillage

Primary tillage for sugarcane involves 5–6 ploughings to break the hard pan, followed by 2–3 plankings and land levelling.

Seed Bed Preparation

In the wet method of seed bed preparation, a pre-soaking irrigation is applied to achieve optimum moisture, followed by 2 ploughings and planking. In the dry method, which follows fallow land, the field is dried and irrigated, followed by 1–2 ploughings and planking.

Sowing Time

Spring sowing takes place in February and March, with the crop maturing in February. Autumn sowing takes place in August and September, with the crop maturing in November or December.

Seed Rate

The recommended seed rate for sugarcane is 75,000 to 100,000 setts per hectare, equivalent to about 200–250 maunds per hectare, or 30–40 marlas per hectare for a normal crop.

Sowing Methods

Sugarcane can be sown using flat sowing, ridge sowing, or pit planting. Row arrangements include single row (60 cm spacing), double row (90 cm and 30 cm spacing), and triple row (120 cm and 30 cm spacing) patterns.

Fertilizer

The recommended fertilizer dose for sugarcane is 200:100-150:100 kg per hectare of NPK. One-third of the nitrogen is applied along with the full amount of phosphorus and potassium at sowing. The remaining nitrogen is split, with one-third applied at tillering, about two months after sowing, and the final one-third applied at earthing up, before the start of the rainy season.

Irrigation

Sugarcane sown in March requires 16–18 irrigations, while sugarcane sown in September requires 20–22 irrigations. The irrigation method used depends on water availability and rainfall.

Weeds

Weed management in sugarcane relies on cultural control, chemical control, and biological control, applied through pre-emergence and post-emergence methods.

Insect Pests

Borer attack can be severe in sugarcane. Management includes the use of systemic granular insecticides when the crop is 1–2 feet high, as well as biological control agents.

Diseases

Sugarcane is affected by diseases such as red rot and whip smut. These are managed through seed treatment and the use of resistant varieties.

Harvesting

Sugarcane sown in March is harvested in January, while sugarcane sown in September is harvested in October or November. Harvesting operations include stripping, trashing, and topping.

Sugarcane Varieties by Province

Punjab: Approved varieties include CP 77/400, CP 43/33, CPF 237, CPF 243, CPF 247, SPF 231, SPF 234, SPF 245, SPF 246, SPF 247, HSF 240, and COJ 284. Prohibited varieties in Punjab include CO 1148 (Indian), Triton, COL 29, COL 44, COL 54, BL 4, L 116, L 118, BF 162, and SPF 238.

Sindh: Approved varieties include Thattha 10, Larkana 1, Larkana 2004, CPF 237, L 126, HS 2, HS 4, HS 12, and B 129 (Gulabi 95). Prohibited varieties in Sindh include COL 29, COL 54, NIA 98, and Disco 843/60.

KPK: Approved varieties include CP 77/400, Mardan 92, Mardan 93, Bannu 1, SPSG 394, MSP 421, and CP 72-2086. Prohibited varieties in KPK include COS 245, COS 321, NOC 310, and NS 98.